Gaurav Kumar Verma

Software Engineer · AI/ML & LLM Engineer · Competitive Programmer

gaurav.k.verma23@gmail.com +91 9267980305 Faridabad, Haryana, India Portfolio LinkedIn Github

PROFESSIONAL SUMMARY

Full-stack and AI/ML engineer who ships production systems end-to-end: built a Clean Architecture Agentic RAG platform (FastAPI, LangChain, ChromaDB, LLM) during a Paytm internship, trained a 98.66%-accuracy brain tumor classifier (EfficientNetV2-S) with Grad-CAM interpretability, and shipped a real-time AQI forecasting API (FastAPI + Scikit-learn) serving geospatial visualizations across Delhi wards. Contributed to OcuScan AI, a deep-learning eye disease classification model built for Sankara Nethralaya Hospital, and won a Certificate of Excellence at Hackoverflow 9.0 for a multi-role healthcare dashboard. Comfortable owning a system from schema design to deployed frontend.

SKILLS

Programming Languages
C, C++, Python, Java, HTML, CSS, JavaScript, TypeScript, SQL

Frontend Developer
React.js, Vite, Tailwind CSS, Responsive Web Design, Next.js, Figma, Canva

Backend Developer
FastAPI, REST APIs, Node.js, Firebase, Supabase, RDBMS, NoSQL, Authentication & Authorization, Testing, Containerization

CS Fundamentals
Data Structures & Algorithms, OOPs, System Design, DBMS, Computer Networks, OS

AI/ML Engineering
Model Evaluation, Scikit-learn, PyTorch, TensorFlow, Ollama, Model Design, NLP, Hyperparameter Tuning

Data Analytics
Numpy, Pandas, Time-series Forecasting, Data Visualization, Data Cleaning, Statistics

Tools & Platforms
Shell, Amazon Web Services (AWS), Git, Github, Vercel, Railway, Render, Docker, VS Code, Anaconda

LLM Engineering
Langchain & Langgraph, Prompt & Context Engineering, Vector Databases, Vector Embeddings, Retrieval-Augmented Generation (RAG), AI Agents & Orchestration

Experience

FusionHawk — Remote
Software Developer Intern — 02/06/2026-18/06/2026

- Built an AI-assisted anterior segment ocular disease classifier by fine-tuning an EfficientNetB0 backbone on a curated dataset of 319 clinical images, achieving a 0.787 Macro AUC-ROC score.
- Implemented anti-overfitting techniques — label smoothing, MixUp augmentation, and 5-fold cross-validation — to maximize diagnostic reliability on rare ocular surface conditions.
- Packaged and deployed the inference model via a multi-screen Streamlit application within a 3.57GB Docker container, featuring real-time Grad-CAM explainability and condition-specific warning banners.

Paytm Payment Services Limited — Noida, India
Intern — 18/06/2026-24/07/2026

- Architected a modular, provider-agnostic Agentic RAG platform using FastAPI, LangChain, React, and ChromaDB, implementing deterministic tool routing, hybrid retrieval (Dense + BM25 + RRF), cross-encoder reranking, query rewriting, and document summarization for PDF knowledge bases up to 2,500 pages.
- Optimized LLM inference with Groq API and Server-Sent Events (SSE), achieving 464.5 ms TTFT and 748.2 ms average response latency while generating grounded, citation-backed responses.
- Engineered a production-style architecture using Strategy, Factory, and Registry patterns with pluggable AI providers, alongside an offline retrieval evaluation framework and 400+ automated tests (69.93% coverage) to validate retrieval quality and system reliability.

Projects

ATLAS (Almost Thinking Local AI System) 2026

Python · Ollama (local LLM) · NLP/Regex Pipeline · Desktop APIs

- Developed a privacy-first, offline desktop AI assistant constrained to a 6GB VRAM environment, routing 25–35 daily voice commands using faster-whisper STT and dynamically selected Ollama models.
- Reduced trivial query latency by 2,029.8ms by designing a regex pre-classifier gate that lets common desktop commands bypass LLM inference entirely, saving GPU cycles.
- Architected a hardware-validated VRAM-tiering benchmark harness measuring model selection against forced-load baselines, producing an avoidance-focused tiering strategy that mitigates Out-of-Memory (OOM) risk.

NeuroVision AI 2026

PyTorch · EfficientNetV2-S · Grad-CAM · Streamlit

- Engineered a production-grade 4-class brain tumor classification system using PyTorch and an EfficientNetV2-S backbone, training on 15,415 MRI scans to achieve 98.66% accuracy.
- Secured a Top 5 finish at the CODEAI Hackathon 2026 by solo-developing the end-to-end pipeline, including a 5-fold cross-validation strategy and Mixed Precision (AMP) training for optimal GPU throughput.
- Deployed a clinical-decision support web application via Streamlit delivering real-time predictions, confidence scores, and custom Grad-CAM explainability heatmaps for tumor localization.

HealthSync 2025

React · Vite · Tailwind · Redux Toolkit · Tesseract.js · Vercel

- Awarded Certificate of Excellence at Hackoverflow 9.0 for co-developing a full-stack post-discharge healthcare platform connecting hospitals and patients, built with React, Vite, and Tailwind CSS.
- Integrated AI-powered OCR via Tesseract.js and the Gemini API, digitizing complex medical prescriptions with an average processing time of 10.58 seconds.
- Engineered backend LLM workflows to parse lab data and vitals in real time, generating predictive emergency alerts and personalized medication adherence schedules.

VOLUNTEER

Technical Head - National Service Scheme (NSS), JIIT Noida 10/2025 – Current

Noida, India

- Led technical operations for community outreach events promoting environmental and social awareness.
- Tutored peers in technical subjects; produced informational content for community distribution.

EDUCATION

Jaypee Institute of Information Technology Bachelor of Technology - BTech • 7.78

Computer Science and Engineering Noida, India • 2024-2028

Achieved 5th place in CodeClash, a campus-wide competitive programming contest conducted on HackerRank.

St. Joseph's Convent School Higher Secondary School • 93

Non Medical Faridabad, India • 2009-2024

CERTIFICATIONS

Certificate of Excellence in Hackoverflow 9.0

Cisco Certified Networking Basics

Certificate of Appreciation in Paranox 2.0

Certificate of Appreciation in HackVision

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

Deloitte Data Analytics Job Simulation